

1 Solution — GIAM 7.3

Problem 1

1.1 Problem

The statement that there are two non-bald New Yorkers with the same number of hairs on their heads requires some careful estimates to justify it. Please justify it.

1.2 Pigeonhole Framing

Let N denote the set of *non-bald* New Yorkers — residents of New York City whose head bears at least one hair. For each $p \in N$, write $h(p) \in \mathbb{Z}_{\geq 1}$ for the (positive) number of hairs on p 's scalp. The function

$$h: N \longrightarrow \{1, 2, 3, \dots, H\}$$

is then well-defined for any integer H that exceeds the hair count of every human alive.

The pigeonhole principle (Theorem 7.3.1) says: *if $|\text{Dom}(h)| > |\text{Rng}(h)|$, then h is not injective.* Concretely, if we can show

$$|N| > H,$$

then two distinct non-bald New Yorkers $p \neq q$ must satisfy $h(p) = h(q)$ — exactly the claim.

The task therefore reduces to producing two numerical estimates with the right inequality:

- a *generous upper bound* H on the number of hairs on any human head, and
- a *conservative lower bound* on $|N|$, the count of non-bald New Yorkers,

and then checking $|N| > H$.

Why “non-bald” matters. Without restricting to non-bald people, every fully bald New Yorker shares the hair count 0 with every other fully bald New Yorker, and

the claim collapses to a triviality. The non-bald restriction makes the problem honest: we have to find a collision among positive hair counts.

1.3 Step 1 — Upper Bound H on Hair Count

Two independent factors bound the number of hairs on a human scalp: the area of the scalp, and the density of follicles upon it.

Scalp area. A typical adult scalp covers roughly $600\text{--}700\text{ cm}^2$. As a *very* generous upper bound — covering even the largest heads — take

$$A_{\max} = 1000\text{ cm}^2.$$

(For comparison: a hemisphere of radius 12.6 cm , larger than the human skull, has surface area $2\pi r^2 \approx 998\text{ cm}^2$.)

Hair density. Dermatological references put average scalp hair density around 200 hairs/cm^2 , with the highest measured densities (typically among light blonds) reaching about $300\text{--}400\text{ hairs/cm}^2$. As an inflated upper bound, take

$$\rho_{\max} = 1000\text{ hairs/cm}^2,$$

— a factor of $2.5\text{--}5\times$ over the densest measured scalps.

Multiplying the two generous bounds:

$$\#\text{hairs on any head} \leq A_{\max} \cdot \rho_{\max} = 1000\text{ cm}^2 \cdot 1000 \frac{\text{hairs}}{\text{cm}^2} = 10^6.$$

We take

$$H = 10^6 = 1,000,000$$

as our pigeonhole count, with substantial slack: realistic counts run $10^5\text{--}1.5 \times 10^5$, so H is inflated by roughly an order of magnitude over biological reality.

1.4 Step 2 — Lower Bound on $|N|$

Population of NYC. Per recent U.S. Census Bureau estimates, the population of New York City exceeds 8×10^6 . As a conservative lower bound, take

$$|\text{NYC}| \geq 7 \times 10^6.$$

Fraction bald. “Bald” here means *zero hairs on the scalp* — a strict criterion. Children are essentially never bald, female-pattern baldness rarely produces zero hairs, and even severe male-pattern baldness usually leaves a fringe of follicles intact. A reasonable upper estimate on the fraction of New Yorkers with literally no scalp hair is well under **0.20**.

Even taking the *absurdly* loose upper estimate that **half** of New Yorkers are completely bald:

$$|N| \geq \frac{1}{2} \cdot 7 \times 10^6 = 3.5 \times 10^6.$$

With a realistic fraction-bald of **0.10**, the lower bound is much larger:

$$|N| \geq 0.9 \cdot 7 \times 10^6 = 6.3 \times 10^6.$$

Either way,

$$\boxed{|N| \geq 3.5 \times 10^6}$$

is safe.

1.5 Step 3 — Apply the Pigeonhole Principle

Combining the bounds:

$$|N| \geq 3.5 \times 10^6 > 10^6 = H.$$

So $h: N \rightarrow \{1, 2, \dots, H\}$ has $|\text{Dom}(h)| > |\text{Rng}(h)|$. By Theorem 7.3.1, h is not injective: there exist distinct $p, q \in N$ with

$$h(p) = h(q).$$

That is, **two non-bald New Yorkers have exactly the same number of hairs on their heads.** ■

1.6 Robustness of the Estimates

The pigeonhole inequality $|N| > H$ holds by a factor of at least $3.5\times$ — comfortable enough to survive much sloppier bounds. Below, “realistic” is the typical biological/demographic value; “used here” is the inflated bound that still gives the argument.

Quantity	Realistic value	Bound used here	Slack
NYC population	$\approx 8.3 \times 10^6$	$\geq 7 \times 10^6$	$\sim 1.2\times$
Fraction bald	$\lesssim 0.10$	≤ 0.50	$\sim 5\times$
Max scalp area	$\lesssim 700 \text{ cm}^2$	$\leq 1000 \text{ cm}^2$	$\sim 1.4\times$
Max hair density	$\lesssim 400/\text{cm}^2$	$\leq 1000/\text{cm}^2$	$\sim 2.5\times$
Realistic max hair count	$\lesssim 2 \times 10^5$	$H = 10^6$	$\sim 5\times$
Non-bald NYC (\$	N	\$)	$\approx 7.5 \times 10^6$
Pigeonhole margin \$	N	/H \$	≈ 37

Table 1.1

Every individual factor could be off by $2\times$ or more and the conclusion would survive. The argument is not merely “true under the stated bounds” — it is *robustly* true.

1.7 A Sharper Conclusion via the Strong Pigeonhole

The *strong* form of the pigeonhole principle (Theorem 7.3.2) says: if $|N|$ objects are distributed among H pigeonholes of equal capacity m , and $|N| > H(m - 1)$, then some pigeonhole contains at least m objects. Equivalently, *some* hair count is shared by at least

$$m = \left\lceil \frac{|N|}{H} \right\rceil$$

different non-bald New Yorkers.

- **Inflated bounds** ($|N| \geq 3.5 \times 10^6$, $H = 10^6$): some hair count is shared by at least $\lceil 3.5 \rceil = 4$ non-bald New Yorkers.

- **Realistic bounds** ($|N| \approx 7.5 \times 10^6$, $H \approx 2 \times 10^5$): some hair count is shared by at least $\lceil 37.5 \rceil = 38$ non-bald New Yorkers — and by averaging, the *typical* hair count is shared by roughly 37.

So the conclusion is not just “a coincidence exists somewhere” — *every* hair count in the natural range is shared by dozens of New Yorkers.

1.8 Remark — The Structure of the Argument

The proof has a familiar shape for pigeonhole problems:

1. **Identify the function.** Map *pigeons* (non-bald New Yorkers) to *pigeonholes* (positive integers up to H) via the natural counting map.
2. **Bound the domain from below.** A lower bound on $|N|$ — from population data and a generous bald-fraction.
3. **Bound the codomain from above.** An upper bound on H — from scalp anatomy.
4. **Verify** $|\text{Dom}| > |\text{Rng}|$. A single arithmetic comparison.

What makes this particular instance worth examining is *Step 3*: it requires biological knowledge external to combinatorics, and one has to argue that the bound is finite *at all* before bounding it. (A priori, one could imagine humans with arbitrarily many hairs — the pigeonhole then has no codomain bound, and the argument fails.) The argument’s substance lies in convincing the reader that hair counts are bounded by *some* number much smaller than the population of NYC.

1.9 Remark — Why “Careful Estimates”

The book asks for *careful* estimates because the argument has two failure modes:

- **Codomain bound too loose.** If one allows, say, $H = 10^{10}$ (“who knows how many hairs?”), the inequality $|N| > H$ fails and the proof breaks. Care is needed to *defend* the upper bound on hair count — i.e., to explain *why* a human head cannot harbor 10^7 hairs.
- **Domain bound too tight.** If one underestimates $|N|$ — e.g., by claiming “95% of people are bald” — the inequality again fails. Care is needed in defining “bald” strictly (literally zero hairs) and recognizing that this is rare.

A careless treatment fails on either flank. The estimates above are *careful* in that each side is inflated/deflated *away* from the desired inequality, so the inequality holds even under hostile reinterpretation of the numbers.